DOD-P-63977A(AR)
7 AUGUST 1986
SUPERSEDING
DOD-P-63977(AR)
14 May 1982

MILITARY SPECIFICATION

PROJECTILE 30MM, TARGET PRACTICE AND TARGET PRACTICE ALTERNATE DESIGN FOR CARTRIDGE, 30MM, M788

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts, equipment, requirements, quality assurance provisions and packaging for the target practice projectile for use in the 30mm M788 TP cartridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, listed in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-A-2550 - Ammunition, General Specification for
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specifications for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1169- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9349343 - Body Assembly, Projecile, 30MM, TP M788
9399639 - Projectile, 30mm, TP, M788 - Painted Assembly
11825947 - Cartridge, 30mm, M788 TP
11825948 - Projectile, 30mm, Target Practice M788

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific aquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DoDISS specified in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-B117 - Method of Test for Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General.

3.1.1 Materials. All materials shall be in accordance with the applicable drawings, specifications and standards.

3.1.2 Projectile. The projectile shall comply with all requirements of Drawing (dwg.) 9399639 or 11825948, as applicable, and with all requirements in applicable specifications.

3.2 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without prior approval (see 6.7).

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.6).

3.4 Banding. The rotating band blank and the rotating band seat shall be clean and dry at the time of banding. Shot or sandblasting shall not be used to clean the band seat after knurling. The rotating band blank shall fill the band seat.

3.5 Soundness. The projectile bodies shall be examined for metal discontinuities by eddy current or ultrasonic inspection. A discontinuity in excess of .25mm deep by 6.35mm in length shall be classified as a defect.

3.6 Protective coating.

3.6.1 Phosphate coating. The pretreatment process used shall be in accordance with the applicable requirements of Dwg. 11825948 or dwg. 9399639. The phosphate coating shall be in accordance with the requirements of TT-C-490 for appearance and minimum weight, as applicable to the type of phosphate coating. Test specimens (panels) shall be in accordance with the requirements of TT-C-490.

3.6.2 Paint coating. The paint coating shall be in accordance with applicable requirements of Dwg. 9349343 or 9399639 exterior paint adhesion shall be in accordance with applicable requirements of TT-C-490.

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3.6.3 Corrosion resistance. The painted projectile body (excluding the nose and aft of the rotating band) shall show no evidence of corrosion (red rust) after exposure to a 5 percent solution of atomized salt spray for 24 hours.

3.7 Metal parts security. The projectile, when assembled into a test cartridge loaded in accordance with the requirements of dwg. 11825947 shall not separate or break-up.

3.8 Workmanship. The requirements for workmanship are specified by applicable drawings, referenced specifications and the following.

3.8.1 Processing defects. The projectile shall be free of defects such as, dents, deep scratches, cuts, sharp edges, burrs, metal flaws, excessive surface roughness and other forms of substandard workmanship.

3.8.2 Cleanliness. The projectile shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. References shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection (Table I).

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as disignated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated, for either design as specified in the contract:

Part Description	Drawing	Quantity
Blank, Rotating Band for Projectile, 30mm	11825929	32
Nose, Projectile	11825905	32
Projectile Body - TP	11825950	32
Body & Band Assembly TP (Unpainted)	11825949	125
Body Assembly-TP (Unpainted)	11825949	32
Projectile, 30mm - TP	9349343	432
Standard Test Panels (see TT-C-490)	11825948	3

Alternate Design

Part Description	Drawing	Quantity
Blank, Rotating Band for Projectile, 30mm	11825929	32
Nose, Projectile	9399643	32
Projectile Body - TP	9399642	32
Body & Band Assembly TP	9399641	125
Projectile Assembly (Unpainted)	9399640	125
Projectile, 30mm - TP (painted)	9399639	432
Standard Test Panels (see TT-C-490)		3

4.3.2 Inspection to be performed. See MIL-A-48078 and Table IA or Table IB as applicable.

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4.3.2.1 First article inspection. Prior to submission of the first article sample to the Government, the contractor shall inspect all components and assemblies for all contract, drawings, and specifications requirements. The contractor's inspection shall be subjected to verification by the Government representative.

4.3.3 Rejection. See MIL-A-48078 and the following:

4.3.3.1 First article sample failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample dissapproval.

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TABLE IA. First article inspection

CLASSIFICATION OF DEFECTS & TESTS DOD-P-63977A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Projectile and Components				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rotating Band (Dwg. 11825922)				
	Examination for defects	32		3.1	4.4.2.1
	Nose, Projectile (Dwg. 11825905)				
	Examination for defects	32		3.1	4.4.2.2
	Body, Projectile (Dwg. 11825950)				
	Examination for defects	32		3.1	4.4.2.3
	Body and Rotating Band Assy (Dwg. 11823949)				
	Examination for defects	32**		3.1	4.4.2.4
	Band Tightness	125	0-1*	3.4	4.5.1/4.5.1.1 1/
	Body Assembly (Dwg. 9349343)				
	Examination for defects	32		3.1	4.4.2.5
	Projectile Assembly (Dwg. 11825948)				
	Examination for defects	32***		3.1	4.4.2.6
	Soundness	125***	0-1*	3.5	4.5.2 2/
	Paint Adhesion	13***	0-1*	3.6.2	4.5.5
	Corrosion Resistance	32	0-1*	3.6.3	4.5.6
	Metal Parts Security	400	0-1*	3.7	4.5.7 3/
	Test Panels per TT-C-490				
	Phosphate coating	3	0-1*	3.6.1	4.5.4 4/

NOTES:
* Accept-reject criteria.
** After examination these assemblies shall be used for band tightness.
*** After examination these assemblies shall be used for metal parts security.

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TABLE IB. First article inspection

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
	Projectile and Components		See below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rotating Band (Dwg. 11825929)				
	Examination for defects	32		3.1	4.4.2.1
	Nose, Projectile (Dwg. 9399643)				
	Examination for defects	32		3.1	4.4.2.7
	Body, Projectile (Dwg. 9399642)				
	Examination for defects	32		3.1	4.4.2.8
	Body and Rotating Band Assy (Dwg. 9399641)				
	Examination for defects	32**		3.1	4.4.2.9
	Band Tightness	125	0-1*	3.4	4.5.1/4.5.1.1 1/
	Projectile Assembly (Unpainted) (Dwg. 9399640)				
	Examination for defects	32***		3.1	4.4.2.10
	Soundness	125***	0-1*	3.5	4.5.2 2/
	Projectile Assembly (Painted) (Dwg. 9399639)				
	Examination for defects	32**			4.4.2.11
	Paint Adhesion	13***	0-1*	3.6.2	4.5.5, 6.9

NOTES:

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TABLE IB. First article inspection

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Projectile and Components				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Corrosion Resistance	32	0-1*	3.6.3	4.5.6
	Metal Parts Security	400	0-1*	3.7	4.5.7 3/
	Test Panels per TT-C-490				
	Phosphate Coating	3	0-1*	3.6.1	4.5.4 4/

NOTES:

* Accept-reject criteria.

** After examination these assemblies shall be used for band tightness.

*** After examination these assemblies shall be used for metal parts security.

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1/ Projectile assemblies shall be sectioned through the longitudinal axis to expose the band fill in the undercut areas. Each undercut area so exposed shall as a minimum be 50% filled in by the rotating band material.

2/ Rejection of one or more units of the sample by the flaw detector in the test for body metal soundness shall be cause for rejection of the lot.

3/ The occurrence of one or more of the following casualities, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s) shall be cause for rejection of the lot, except as otherwise provided:

a. Body breakup.

b. Loss of nose.

c. Complete rotating band separation. This is defined as detachment of all of the band having band seat knurling imprint.

d. Partial rotating band separation. This is defined as detachment of part of the band having band seat knurling imprint.

e. The following acceptance criteria shall apply to thrown band fragmentations having no evidence of band seat knurling imprint:

(1) The lot will be accepted if two or less assemblies throw band fragmentations having length not greater than 13mm.

(2) The lot will be rejected if any assembly throws a band fragmentations having length greater than 13mm (indicative of deep-seated band lamination).

(3) The lot will not be penalized for band particles thrown as a result of normal band fringing or slivering.

4/ Failure of phosphate coating on the standard panels or sample body assemblies to comply with the applicable requirements shall be cause for rejection of all projectiles phosphated since the preceding test.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination for defects. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects, and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 11825929
4.4.2.1	Rotating Band (for either design)		NEXT HIGHER ASSEMBLY 11825949, 9399641

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major		See 4.4.2			
101.	Defective metal, including lamination		.25%	3.1.1	Visual
102.	Incomplete or deformed		.25%	3.1.1	Visual 1/
Minor					
201.	Width, maximum		.40%	3.1.1	Gage
202.	Workmanship		.65%	3.8 & 3.4	Visual

NOTES: 1/ A rotating band blank which is deformed or judged to lack sufficient metal in any part of its width or thickness to provide for complete band seat fill and minimum rotating band profile dimensions shall be classed defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.2	Nose, Projectile	1	1	11825905
				NEXT HIGHER ASSEMBLY 11825948

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major		See 4.4.2			
101.	Base diameter min. and max.		.25%	3.1	Gage
102.	Step diameter min. and max.		.25%	3.1	Gage
103.	Profile, Ogive		.25%	3.1	Profile Comparator Chart (Complete Ogive)
Minor					
201.	Workmanship		.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 11825950
4.4.2.3	Body, Projectile		NEXT HIGHER ASSEMBLY 11825949

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major		See 4.4.2			
101.	Outside dia, base min.		.25%	3.1	Gage
102.	Inside dia, base, max.		.25%	3.1	Gage
103.	Inside dia, Middle, Max.		.25%	3.1	Gage
104.	Length, mouth to rear base of rotating band seat		.25%	3.1	Gage
105.	Diameter, rotating band seat		.25%	3.1	Gage
106.	Inside dia, mouth		.25%	3.1	Gage
107.	Knurling missing or incomplete		.25%	3.1	Visual
108.	Crimp groove profile		.25%	3.1	Comparator
109.	Foreign matter in rotating band seat		.25%	3.1	Visual (see 3.4)
110.	Base radius		.25%	3.1	Compatator
111.	Runout inside cavity dias. (3) with datum surface		.25%	3.1	Gage
112.	Presence of mouth relief and chamfer		.25%	3.1	Visual
113.	Length from base to rear of band seat		.25%	3.1	Comparator
114.	Weight		.25%	3.1	Scale/Balance
Minor					
201.	Width, rotating band seat		.40%	3.1	Gage
202.	Profile mouth dia and mouth break		.40%	3.1	Comparator
203.	Finish, improper		.40%	3.1	Visual
204.	Workmanship		.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 11825949
4.4.2.4	Body and Rotating Band Assembly 1/				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major		See			
101.	Dia, bourrelet, min.	4.4.2	.25%	3.1	Gage
102.	Dia, rotating band		.25%	3.1	Gage
103.	Length, mouth to rear base of rotating band seat		.25%	3.1	Gage
104.	Runout rotating band w/datum surface		.25%	3.1	Gage
105.	Void, junction rotating band and body		.25%	3.1	4.5.1/Gage
106.	Band tightness	20	ACC-0 REJ-1	3.1	4.5.1 2/
Minor					
201.	Depth and width of undercut, front and rear of rotating band, max	See 4.4.2	.40%	3.1	Profile comparator chart for (min 201-202)
202.	Width, rotating band		.40%	3.1	
203.	Corrosion		.40%	3.1	Visual
204.	Workmanship		.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 11825949
4.4.2.4	Body and Rotating Band Assembly 1/				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	1/ If crimp groove configuration is machined at this operation refer to para 4.4.2.3 CD 108, 110 and 113. 2/ Projectile assemblies shall be sectioned through the longitudinal axis to expose the band fill in the undercut areas. Each undercut area so exposed shall as a minimum, be 50% filled in by the rotating band material.				

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9349343
4.4.2.5	Body, Assembly, Projectile				NEXT HIGHER ASSEMBLY 11825948
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Dia rotating band max.		.25%	3.1	Gage
102.	Foreign matter in mouth ID cavity		.25%	3.1	Visual
103.	Dia bourrelet max		.25%	3.1	Gage
104.	Dia base max		.25%	3.1	Gage
105.	Phosphate coating (3 standard panels)	3	.25%	3.1	4.5.4 1/
106.	Paint adhesion projectile painted	2	.25%	3.1	4.5.5, 6.9
Minor					
201.	Exterior paint inadequate		.40%	3.1	Visual
202.	Foreign matter on exterior		.40%	3.1	Visual
203.	Workmanship		.65%	3.8	Visual

NOTES: 1/ Failure of the phosphate coating on the standard panels or sample body assemblies to comply with the applicable requirements shall be cause for rejection of all projectiles phosphated since the preceding test.

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AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Projectile Assembly		11825948
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Soundness, projectile (dwg. 11825948)	All	100%	3.5	4.5.2 1/
Major					
101.	Nose, gap	See 4.4.2	.25%	3.1	Gage/Visual
102.	Metal parts security	75	Acc-0 Rej-1	3.7	4.5.7 1/
Minor					
201.	Marking incorrect, incomplete, illegible or missing	See 4.4.2	.40%	3.1	Visual
202.	Damaged rotating band		.40%	3.1	Visual
203.	Loose nose		.40%	3.1	Manual 2/
204.	Workmanship		.65%	3.8	Visual

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1/ Rejection of one or more units of the sample by the flaw detector in the test for body metal soundness shall be cause for rejection of the lot.

2/ The occurrence of one or more of the following casualities, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s) shall be cause for rejection of the lot, except as otherwise provided:

a. Body breakup.

b. Loss of nose.

c. Complete rotating band separation. This is defined as detachment of all of the band having band seat knurling imprint.

d. Partial rotating band separation. This is defined as detachment of part of the band having band seat knurling imprint.

e. The following acceptance criteria shall apply to thrown band fragmentations having no evidence of band seat knurling imprint:

(1) The lot will be accepted if two or less assemblies throw band fragmentations having length not greater than 13mm.

(2) The lot will be rejected if any assembly throws a band fragmentations having length greater than 13mm (indicative of deep-seated band lamination).

(3) The lot will not be penalized for band particles thrown as a result of normal band fringing or slivering.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9399643
4.4.2.7	Nose Projectile Alternate design				NEXT HIGHER ASSEMBLY 9399640
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Profile ogive	See 4.4.2	.25%	3.1	Profile comparator chart (complete ogive)
Minor					
201.	Base diameter		.40%	3.1	Gage
202.	Base Length		.40%	3.1	Gage
203.	Surface finish of ogive		.65%	3.1	Visual Standards
204.	Knurl missing		.65%	3.2	Visual
205.	Workmanship		.65%	3.8	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9399642
4.4.2.8	Body Projectile Alternate design				NEXT HIGHER ASSEMBLY 9399641
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major		See			
101.	Outside diameter base, min	4.4.2	.25%	3.1	Gage
102.	Inside dia, base, max		.25%	3.1	Gage
103.	Inside dia, middle, max		.25%	3.1	Gage
104.	Length, mouth to rear base of rotating band seat 1/		.25%	3.1	Gage
105.	Diameter, rotating band seat		.25%	3.1	Gage
106.	Knurling missing or incomplete		.25%	3.1	Visual
107.	Crimp groove profile		.25%	3.1	Comparator
108.	Foreign matter in rotating band seat		.25%	3.1	Visual (see 3.4)
109.	Base radius		.25%	3.1	Comparator
110.	Runout inside cavity diameters (3) with datum surface		.25%	3.1	Gage
111.	Length from base to rear of band seat		.25%	3.1	Comparator
112.	Weight		.25%	3.1	Scale/Balance
Minor					
201.	Width, rotating band seat		.40%	3.1	Gage
202.	Finish improper		.40%	3.1	Visual
203.	Workmanship		.65%	3.8	Visual

NOTES:

1/ Verification of band seat may be made by inspecting a minimum of two units at start of each shift and at a minimum each four hours of production or tool change.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9399641
4.4.2.9	Body and Rotating Band Assembly Alternate design 1/				NEXT HIGHER ASSEMBLY 9399640
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major		See			
101.	Dia, rotating band	4.4.2	.25%	3.1	Gage
102.	True position of band w/datum surface		.25%	3.1	Gage
103.	Gap, Junction rotating band and body		.25%	3.1	4.5.1.1
104.	Band tightness	20	Acc-0 Rej-1		4.5.1 2/
105.	Length, rear to band to base	See 4.4.2	.25%	3.1	Gage
Minor					
201.	Depth and width of undercut, front and rear of rotating band, max		.40%	3.1	Profile comparator chart for minor 201 and 202
202.	Width, rotating band		.40%	3.1	
203.	Workmanship		.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63977A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9399641
4.4.2.9	Body and Rotating Band Assembly Alternate design 1/				NEXT HIGHER ASSEMBLY 9399640
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	1/ If crimp groove configuration is machined at this operation refer to para 4.4.2.8 CD 107, 109 and 111. 2/ Projectile assemblies shall be sectioned through the longitudinal axis to expose the band fill in the undercut areas. Each undercut area so exposed shall as a minimum, be 50% filled in by the rotating band material.				

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Projectile Assembly Alternate design		9399640
			NEXT HIGHER ASSEMBLY 9399639

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Soundness, projectile	All	100%	3.5	4.5.2 2/
Major					
101.	Nose gap	See	.25%	3.1	Gage
102.	Diameter of bourrelet	4.4.2	.25%	3.1	Gage
103.	Nose runout		.25%	3.1	Gage
104.	Length, rear of band to nose tip max		.25%	3.1	Gage
105.	Body ogive protruding above flush with nose ogive		.25%	3.1	Visual/Manual
Minor					
201.	Loose nose		.40%	3.1	Manual*
202.	Profile ogive		.65%	3.1	Comparator chart
203.	Workmanship		.65%	3.8	Visual

NOTES:

* If the nose moves with respect to the projectile body under thumb and finger pressure applied in the lateral rotational or axial direction, the projectile shall be classified defective.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63977A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER 9399639
4.4.2.11	Projectile Assembly, painted Alternate design		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major		See			
101.	Dia, rotating band, max	4.4.2	.25%	3.1	Gage/Visual
102.	Dia, bourrelet, max		.25%	3.1	Gage
103.	Dia, base, max		.25%	3.1	Gage
104.	Phosphate coating (3 standard panels)	3	Acc-0 Rej-1	3.6.1	4.5.4. 1/
105.	Paint adhesion to projectile	2	Acc-0 Rej-1	3.6.2	4.5.5, 6.9
106.	Metal parts security	75	Acc-0 Rej-1	3.7	4.5.7 2/
Minor					
201.	Exterior paint inadequate		.40%	3.1	Visual
202.	Foreign matter on exterior		.40%	3.1	Visual
203.	Workmanship		.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Projectile Assembly, painted Alternate design	2 OF 2	9399639 NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	1/ Failure of phosphate coating on the standard panels or sample body assemblies to comply with the applicable requirements shall be cause for rejection of all projectiles phosphated since the proceeding test 2/ The occurance of one or more of the following casualties, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s) shall be cause for rejection of the lot, except as otherwise provided. a. Body breakup. b. Loss of nose. c. Complete rotating band separation. This is defined as detachment of all of the band having band seat knurling imprint. d. Partial rotating band separation. This is defined as detachment of part of the band having band seat knurling imprint. e. The following acceptance criteria shall apply to thrown band fragmentations having no evidence of band seat knurling imprint: (1) The lot will be accepted if two or less assemblies throw band fragmentations having length not greater than 13mm. (2) The lot will be rejected if any assembly throws a band fragmentations having length greater than 13mm (indicative of deep-seated band lamination). (3) The lot will not be penalized for band particles thrown as a result of normal band fringing or slivering.				

NOTES:

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4.4.3 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Gap. A .254mm feeler gage shall be used to inspect the gap between the rotating band and projectile body both forward and aft for 360° degrees. The feeler gage shall not enter the gap at any location.

4.5.1.2 Band tightness. After sectioning along the longitudinal axis, one half of the sectioned projectile body shall be examined for rotating band fill in the undercut area. Failure of either of the exposed sections to comply with the 50% fill requirements shall be considered a defect.

4.5.2 Soundness. The procedures for conducting the soundness inspection shall be prepared by the contractor and submitted to the contracting offficer for approval. The procedure shall make provisions for the use of defect standards.

4.5.2.1 Defect standards. As a minimum, the defect standards used in the soundness test will be cycled through the soundness test equipment at the beginning and end of every shift. Failure of the test equipment to reject the defect standards shall be cause for rejection of assemblies inspection for soundness since the preceding test of the defect standards.

4.5.3 Automatic inspection equipment calibration verification. The defect standards for soundness shall be cycled through the soundness inspection equipment at the start and completion of each shift as well as after any adjustment or shutdown of the soundness inspected equipment. Failure of the equipment to reject the defect standards shall be cause for rejection of assemblies inspected for soundness since the preceding test of the defect standards.

4.5.4 Phosphate coating. The method of test shall be as prescribed by TT-C-490 for the type of coating used (see dwg. 11825948 or Dwg. 9399639). The prescribed panels shall be processed, with the projectiles to be represented, at the start of daily production and at 4-hour intervals, maximum, thereafter. Appearance and weight of the coating shall be recorded.

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4.5.5 Paint adhesion. The method of test shall be prescribed by TT-C-490. The test shall be made on the painted bourrelet of each unit of the sample forward of the rotating band. Visible exposure of underlying phosphate coating or ferrous metal of any unit of the sample, as a result of the test, shall be recorded as a defect.

4.5.6 Corrosion resistance. The method of test shall be prescribed in ASTM-B117. The salt spray solution and exposure time shall be as specified in 3.6.3.

4.5.7 Metal parts security. Sample cartridges shall be fired in 25 round bursts at ambient temperature.

4.5.7.1 Test gun. Unless otherwise directed by the contracting officer, the metal parts security firing tests shall be conducted in the 30mm, M230 gun.

5. PRESERVATION, PACKAGING AND PACKING

5.1 Packaging, packing and marking inspection. Inspection for packaging, packing and container marking shall be in accordance with the quality assurance provisions of MIL-STD-1169.

5.2 Packaging and packing, level C. Each lot of projectiles shall be packed using best commercial packaging and packing as approved by the procuring agency.

5.3 Marking and labeling. Marking or labeling of packed cartons or containers shall comly with the applicable requirements of MIL-STD-1169.

5.4 Shipping. When shipments of more than one lot are made at one time, such as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The specification covers assembling, metal parts and packing for Projectile 30mm, Target Practice, M788.

6.2. Ordering data. See MIL-A-48078.

6.3. Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDEC, ATTN: AMSMC-QAF-I(D), Dover, NJ, 07801-5001. Request letter of submittal shall state the contractor, contract number, specification number, item nomenclature and classification of defects or test paragraphs.

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6.4 Submission of results of contractor-conducted examinations and test. Unless otherwise specified by the contracting officer, the contractor shall forward requested records of examinations or tests to Commander, ARDEC, ATTN: AMSMC-QAF-S(D), Dover, NJ, 07801-5001.

6.5 Submission of test data. In addition to the normal distribution or records, when the cartridges are procured by ARDEC, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to Commander, ARDEC, ATTN: AMSMC-QAF-S(D), Dover, NJ, 07801-5001.

6.6 First article. When a first article inspection is required, the contracting officer should include specific instructions in acquisition documents regarding arrangement for examinations, test and approval of the documents for first article.

6.7 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture or an operational change which alters the metallurgical or physical properties of the item of contract.

6.8 Intermediate point inspection. The classification of defects identifies the defect characteristics for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality in the end item.

6.9 Projectile coating defect criteria. Projectiles indicating exposure of bare metal or underlying phosphate shall be considered unsatisfactory if any of the following conditions exist:

a. Any area exceeding 60 square millimeters.

b. More than one area exceeding 30 square millimeters.

c. More than five areas of 15 square millimeters or less.

6.10 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM, Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

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6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issues due to the extensive changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-P-63977A	PROJECTILE 30MM, TARGET PRACTICE AND TARGET PRACTICE ALTERNATE

3a. NAME OF SUBMITTING ORGANIZATION DESIGN FOR CARTRIDGE, 30MM M788

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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